
11. Issue spread determinants in the green bond market: The role of second party reviews and of the Green Bond Principles

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1 INTRODUCTION

This chapter investigates how issuers and investors in the green bond market are benefiting from the adoption of second party reviews and from the development of widely accepted market standards – that is, the Green Bond Principles. The green bond market has experienced exponential growth since inception in mid-2007 (annualized rate of 165 per cent): as of September 2015, it accounts for roughly USD72 billion and, as issuers and investors gain more interest and confidence, it is set to represent a fundamental channel for the financing of green projects in the future.

Fuelled by this remarkable growth trajectory, the green bond market has increasingly attracted attention, and literature regarding this subject has developed at a very fast pace. Importantly, both academic papers and professional reports are contributing to understanding how these instruments are addressing the need for green financing and how investors are reacting to the possibility of investing in sustainable projects through bond securities. In a still-developing scenario, this work positions itself in an unusual way: building on previous findings by KPMG International (2015), it tries to address two very specific research questions by means of a statistical analysis. This is quite rare in the literature given that qualitative and descriptive studies seem to be more common.

The research objective of this work, thus the main contribution to the existing literature, is to quantitatively measure how recent standardization taking place in the market benefits green bond issuers in terms of lower issue spreads.¹ In particular, if there was empirical evidence that green bonds opting into recent standardization measures (i.e., complying with Green Bond Principles guidelines, second party reviews) were able to obtain, *ceteris paribus*, lower issue spread compared with those opting out, an important signal would be captured: that such initiatives do have an impact on issuers' cost of funding and, accordingly, represent a key driver in shaping the future of the green bond market. Accordingly, two research questions are raised: (1) Are green bond issue spreads decreasing in response to the introduction of the Green Bond Principles? (2) Are the issue spreads of green bonds with a second party review lower than those without?

The results of the work are interesting and point in a clear direction: standardization measures have impacted significantly on the issue spreads of green bonds. Indeed, results show that, in response to the introduction of the Green Bond Principles, bonds with a worse credit rating have been able to enter the green bond market and that the green bonds accompanied by a second party review have on average, *ceteris paribus*, a lower issue spread.

The work is organized as follows. Section 2 presents an overview of the relevant literature.

Section 3 provides an overview of the green bond market, touching on the definition of a green bond (subsection 3.1), the development of a market for green bonds (subsection 3.2), and the standardization process occurring in the marketplace (subsections 3.3 and 3.4). The empirical analysis used to address the research questions is presented in section 4: the methodology adopted is described in subsection 4.1, a thorough description of the sample analysed follows in subsection 4.2 and the output of the empirical analysis in subsection 4.3. Section 5 concludes and offers an overview of the main findings.

2 LITERATURE REVIEW

Given that a market for green bonds is still maturing, the literature on the subject is currently in a development phase. Publications are both of a professional nature (e.g., reports written by financial institutions and consulting firms) and academic papers: some of the most relevant work is briefly outlined below.

Joshi (2013) recognized green bonds as an important market opportunity to bring more liquidity and cheaper cost of capital for renewable energy (especially solar) assets. Kidney and Oliver (2014) evidenced how a green bond induces reciprocal collaboration between investors and issuers to increase the financing of green investments. Lindenberg (2014) included green bonds in a group of eight public instruments that can be used to leverage private capital for green investments in developing and emerging countries. Falsen and Johansson (2015) verified that, according to their study and methodology, green bonds deliver significant benefits in terms of company reputation and access to capital. They also suggested that a green bond issuance may have implications on corporate governance by adding another layer to the investment decision process – that is, the environmental dimension.

KPMG International (2015) offered a comprehensive overview of the risks and challenges that the green bond market presents to issuers, particularly those linked to the lack of clear definitions of green use of the bonds' proceeds, of requirements on how the proceeds should be tracked, managed and reported, and of assurance requirements over the information reported. Recent developments occurring in the marketplace (e.g., Green Bond Principles and second party reviews) have been welcomed as positive factors in encouraging investors' confidence about green bonds as well as reducing transaction costs for issuers. In fact, KPMG concluded that bond issuers should not wait for mandatory requirements to emerge before taking a thorough approach in the management of green bond proceeds: doing so would likely build greater confidence with investors, rating agencies and underwriters, thus translating into immediate benefits.

The Climate Bonds Initiative (2015a) estimated that, as of July 2015, approximately 65 per cent of the green bonds in the market are accompanied by a second party review, a figure that testifies to its importance for issuers who want investors to be happy with the credentials of a green bond. In their argumentation, a second party review provides the level of independent assurance that is of particular importance for smaller investors who typically have less resources with which to educate themselves on each potential framework and its underlying 'greenness'. Along the same lines, the Climate Bonds Initiative (2013) argued that, in the current European marketplace, a second party review is more or less a requirement to ensure legitimacy of the issuer even though no requirement or regulations are already in place.

3 AN OVERVIEW OF THE GREEN BOND MARKET

3.1 What is a Green Bond?

A green bond is a bond like any other: a contract by which the issuer borrows money from investors and repays principal and interest according to the terms and conditions set out in the bond's documentation. This means that a green bond investor takes the same credit risk on the issuer's repayment capability that another investor bears on the same issuer's 'uncoloured' bonds. A green bond differs from a normal bond in the following aspects:

- *Use of proceeds.* By definition, green bonds' proceeds are channelled towards projects or activities that promote climate change mitigation, adaptation or other environmental sustainability purposes. The marketplace recognizes as 'green' the following uses of proceeds:
 - renewable energy: solar, wind, hydro, geothermal, biomass, biofuel, bioenergy, waste-to-energy, tidal and other renewable sources;
 - energy smart technologies and energy efficiency: electric and hybrid vehicles, fuel cells, energy storage, digital energy and energy efficiency (buildings, industry, LED and smart lighting);
 - green buildings and infrastructures: new LEED,² Energy Star,³ BREEAM⁴ or other certified building and energy-efficient social infrastructure;
 - agriculture and forestry: forestry management, reforestation, forestation and land-use;
 - other sustainability purposes: climate change adaptation-mitigation, including waste management, clean water and pollution control.
- *Transparency and reporting.* Issuers are required to disclose the projects intended to be financed by the bond's proceeds in order to allow investors to evaluate whether the underlying 'greenness' fits with their investment mission. As part of the disclosure, issuers should outline the process they have used for selecting such projects, the criteria making the projects eligible for benefitting from the green bond's proceeds and, if applicable, the temporary investment of unallocated proceeds. In addition, issuers are recommended to adopt qualitative and quantitative performance indicators – for example, reduction in greenhouse gas emissions due to the projects – as well as seeking independent assurance to provide greater transparency on project evaluation.

3.2 Early Developments and Current Trends

The first green bond was issued in June 2007, when the European Investment Bank launched a five-year AAA EUR600 million bond. The market evolved rapidly and, as depicted in Table 11.1, after the first years (i.e., 2008 to 2012) when issuing green bonds was the prerogative of sovereigns, supranationals and agencies (SSAs) it has now opened to corporates and financials as well. Overall, the green bond market, as of September 2015, accounted for roughly USD72 billion of notional amounts outstanding, which has been mostly denominated in G3⁵ currencies (195 out of 417 bonds, or roughly 75 per cent of the total issued amount) despite other currencies having been active as well, in particular the Swedish krona (SEK), the British pound (GBP) and the Australian dollar (AUD).

3.3 Market Standardization: Green Bond Principles and the Climate Bond Standard

As stated in the original version of 13 January 2014 of the *Green Bond Principles* (International Capital Market Association, 2014): ‘The Green Bond principles are voluntary process guidelines for issuing Green bonds that recommend concrete process and disclosure for issuers which investors, banks, investment banks, underwriters, placement agents and others may use to understand the characteristics of any given Green Bond’. In fact, the Green Bond Principles (GBPs) are intended for broad use by the market to provide issuers with guidance on the key components involved in launching a credible green bond, aid investors by ensuring availability of information necessary to evaluate the environmental impact of their green bond investments, and assist underwriters by moving the market towards greater disclosure, which will facilitate transactions. In order to achieve such goals, the GBPs are built on four pillars:

1. *Use of proceeds.* Proceeds will be addressed to green projects only. According to the GBPs, green projects are those aiming to address key areas of environmental concern such as climate change, natural resources depletion, biodiversity conservation and pollution. The GBPs propose a (not exhaustive) list in an attempt to better categorize the broad areas described above. The list includes: renewable energy, energy efficiency, sustainable waste management, sustainable land use, biodiversity conservation, clean transportation, sustainable water management, climate change adaptation.
2. *Process for project evaluation and selection.* As part of the issuance exercise, GBPs require the issuer to outline the decision-making process adopted to determine the eligibility of the projects to use green bond proceeds. The requirement includes: a process to determine how the projects fit within the eligible categories identified in the GBPs, the environmental sustainability objectives to achieve, and a process in place (if applicable) to quantify those objectives.
3. *Management of proceeds.* The green bond net proceeds should be tracked by the issuer in an appropriate manner – that is, credited to a sub-account, moved to a sub-portfolio or earmarked – and monitored with a formal process. The proceeds should ideally be invested in underlying projects within one year of bond issuance and, in any case, no more than two years after. While green bonds are outstanding, the balance of the tracked proceeds should be periodically reduced by amounts matching eligible green investments undertaken or loan disbursements made during the period. In the case that funds remain uninvested in green projects, the issuer should make known to investors the intended types of temporary investment instruments for the balance of unallocated proceeds. Importantly, GBPs suggest (but do not require) that the management process followed by the issuer be audited by a third party able to verify the internal tracking method for the flow of funds from the green bond proceeds.
4. *Reporting.* Issuers should report on the use of proceeds at least annually via newsletters, website updates or filed financial reports on the specific investments financed through the issuance. A list of projects to which green bond proceeds have been allocated to should – at least – include a brief description of the projects and the amounts disbursed, as well as the expected environmentally sustainable impact, when possible with regard to confidentiality and competitive considerations. Any

Table 11.1 *The green bond market: Yearly issuances and volumes*

Nature of Issuer	Currency/ Rating Class	2007		2008		2009		2010	
		<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)
SSAs		1	669 717 602	1	396 971 169	4	829 571 627	55	2 908 559 673
<i>of which</i>	USD	–	–	–	–	2	480 000 000	2	210 000 000
	IG	–	–	–	–	–	–	1	10 000 000
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	2	480 000 000	1	200 000 000
	EUR	1	669 717 602	–	–	–	–	1	2 232 392
	IG	–	–	–	–	–	–	1	2 232 392
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	1	669 717 602	–	–	–	–	–	–
	JPY	–	–	–	–	–	–	1	1 042 449
	IG	–	–	–	–	–	–	1	1 042 449
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	Other	–	–	1	396 971 169	2	349 571 627	51	2 695 284 832
	IG	–	–	–	–	–	–	15	899 394 556
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	1	396 971 169	2	349 571 627	36	1 795 890 276
Corporates		–	–	–	–	–	–	–	–
<i>of which</i>	USD	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	EUR	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	JPY	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	Other	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
Financials		–	–	–	–	–	–	–	–
<i>of which</i>	USD	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	EUR	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	JPY	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
	Other	–	–	–	–	–	–	–	–
	IG	–	–	–	–	–	–	–	–
	Sub-IG	–	–	–	–	–	–	–	–
	Not rated	–	–	–	–	–	–	–	–
Total		1	669 717 602	1	396 971 169	4	829 571 627	55	2 908 559 673

Note: This table shows the green bonds issued during the period 2007–September 2015 according to the year of issuance, the type of issuer and the currency of issue; the amount issued is also shown. IG = investment grade; YTD = year to date.

2011		2012		2013		2014		2015 YTD	
<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)	<i>N</i>	Issued (USD Equiv.)
31	1092085365	20	2101462066	24	9433416588	56	14505010160	49	6073356570
10	639831000	6	595000000	9	4810000000	17	4333016000	16	3475404000
—	—	1	20000000	8	4260000000	13	3613016000	10	2626087000
—	—	—	—	—	—	—	—	—	—
10	639831000	5	575000000	1	550000000	4	720000000	6	849317000
2	40004465	3	613349704	2	3906686014	9	5234959258	2	703203483
1	22323920	1	390668601	1	3348588012	6	5123339658	2	703203483
—	—	—	—	—	—	—	—	—	—
1	17680545	2	222681103	1	558098002	3	111619600	—	—
—	—	—	—	—	—	1	41697940	—	—
—	—	—	—	—	—	1	41697940	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
19	412249900	11	893112362	13	716730574	29	4895336962	31	1894749087
1	72978833	5	627224732	7	599926256	14	4177767485	7	1498021601
—	—	—	—	—	—	—	—	—	—
18	339271067	6	265887630	6	116804318	15	717569477	24	396727486
—	—	—	—	3	1716722419	46	11940122264	77	13382841506
—	—	—	—	—	—	13	3451800000	48	5759528000
—	—	—	—	—	—	3	1070000000	1	500000000
—	—	—	—	—	—	4	1600000000	5	2850000000
—	—	—	—	—	—	6	781800000	42	2409528000
—	—	—	—	1	1562674406	11	6522156490	17	6160698739
—	—	—	—	1	1562674406	6	5580980020	6	3348588012
—	—	—	—	—	—	2	591583882	6	1964504967
—	—	—	—	—	—	3	349592588	5	847605760
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	2	154048013	22	1966165774	12	1462614767
—	—	—	—	—	—	1	385148657	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	2	154048013	21	1581017117	12	1462614767
—	—	—	—	12	1108225317	27	2071718057	11	3494241655
—	—	—	—	2	552000000	2	269000000	2	1100000000
—	—	—	—	1	500000000	—	—	2	1100000000
—	—	—	—	—	—	—	—	—	—
—	—	—	—	1	520000000	2	269000000	—	—
—	—	—	—	2	334858801	3	1132938944	2	1674294006
—	—	—	—	2	334858801	3	1132938944	1	558098002
—	—	—	—	—	—	—	—	—	—
—	—	—	—	—	—	—	—	1	1116196004
—	—	—	—	3	157576516	5	261654574	—	—
—	—	—	—	—	—	—	—	—	—
—	—	—	—	3	157576516	5	261654574	—	—
—	—	—	—	5	637900000	17	650224539	7	719947649
—	—	—	—	—	—	1	378272053	1	419404446
—	—	—	—	—	—	—	—	—	—
—	—	—	—	5	637900000	16	271952486	6	300543203
31	1092085365	20	2101462066	39	12258364324	129	28516850481	137	22950439731

unallocated proceeds should be reported as well, coupled with a description of the investment instruments in which these proceeds are invested. GBPs also recommend that the issuer reports qualitative and, where applicable, quantitative performance indicators (e.g., annual energy savings in MWh) related to the impact of the specific projects with the same yearly frequency.

The Green Bond Principles are not the only example of increasing standardization and framework development in the green bond market. Indeed, the Climate Bonds Initiative – ‘the only organization in the world working to mobilize the \$100 trillion bond market for climate change solutions’ – promoted a standard for issuing green bonds (Climate Bonds Initiative, 2015b). The Climate Bonds Standard is a tool that allows investors and intermediaries to assess the environmental integrity of green bonds. The prototype of the standard was released in late 2011 and had been used for only five certifications as of August 2015, and a new version (Version 2.0) was published in December 2015 (International Capital Market Association, 2015). The latest version (Version 2.1), launched in 2017, introduces process improvements and an increase in product reach. The range of debt instruments has expanded and there is an inclusion of a ‘programmatic certification option that will streamline the verification process for regular issuers with large portfolios of eligible assets’ (Climate Bonds Initiative, 2017). The standards are multi-sector and are backed by the Climate Bonds Standard Board, which collectively represents USD34 trillion of assets under management.

3.4 Second Party Reviews

As the market continued its development, the need for external assurance regarding the key features of green bonds became urgent. In order to obtain such assurance, issuers have the option to accompany their bond with a second party review offered by an expert consultant with climate expertise (i.e., the reviewer⁶), who provides the issuer and investors with a comprehensive evaluation of the green bond documentation, use of proceeds, compliance with the Green Bond Principles as well as on the robustness and liability of issuers to comply with environmental and social objectives. The review may be kept private at the issuers’ discretion but, as highlighted in International Financing Review (2015), distributing them to the public promotes a higher level of independent assurance that is of particular importance for smaller investors who may have relatively less time and resources to educate themselves on each issuance and its underlying ‘greenness’.

4 EMPIRICAL ANALYSIS

4.1 Methodology and Independent Variables

The empirical analysis builds on previous studies, in particular Gabbi and Sironi (2005), that link a bond’s issue spreads with investors’ perceptions regarding the creditworthiness of the issuer, the bond’s primary and secondary market efficiency and liquidity conditions.

Consequently, the issue spread may be expressed as a function of five main factors: credit rating, bond characteristics, issuer characteristics, market practices and primary market efficiency. Following this reasoning, the research questions were addressed by means of two dummy variables included in the market practices factor – that is, *Reviewer_Dummy* and *GBPs_Dummy*:

$$\text{Issue spread vs Mid-swap}_i = f(\text{Credit rating}_i, \text{Bond characteristics}_i, \text{Issuer characteristics}_i, \text{Market practices}_i, \text{Primary market efficiency}_i) + \varepsilon_i$$

Credit rating A bond's rating is a function of its expected default risk and its recovery rate in the case of default. The rating dummies below should capture the difference in the creditworthiness of the issuers as well as reflect the different risks deriving from bonds' seniority and security structures:

Rating_NR/NA

Dummy variable that takes value 1 if the Bloomberg Composite Rating is missing for the bond under consideration (i.e., the bond is not rated or the rating is not available). The variable is omitted to avoid collinearity in the model and, accordingly, represents our reference in interpreting the other rating dummies.

Rating_HIGHGRADE

Dummy variable that takes value 1 if the Bloomberg Composite Rating of the bond is equal to AAA, AA+, AA or AA– and 0 otherwise. The sign of the coefficient is expected to be negative since a high-grade bond should, *ceteris paribus*, have a lower issue spread than an unrated one. The use of rating as an independent variable is common in the literature and the resulting coefficient is always statistically different from zero (the sign depends on how the variable is constructed); see, for example, Gabbi and Sironi (2005) or Buscaino et al. (2012).

Rating_MEDIUMGRADE

Dummy variable that takes value 1 if the Bloomberg Composite Rating of the bond is equal to A+, A, A–, BBB+, BBB, BBB– and 0 otherwise. Using the same reasoning adopted for *Rating_HIGHGRADE* the sign of the coefficient is expected to be negative.

Rating_SUBIG

Dummy variable that takes value 1 if the bond has a Bloomberg Composite Rating below BBB– and 0 otherwise. By construction, the sign of the coefficient is expected to be positive for a similar reasoning to the one adopted for *Rating_HIGHGRADE* and *Rating_MEDIUMGRADE*.

Bond characteristics The variables below capture the role of some fundamental characteristics of a bond in determining its issue spread, which may not have been accounted for by the credit ratings:

<i>Tenor</i>	This independent variable represents the life of the bonds in years. The expected sign of the coefficient is positive as a longer life to maturity for the bond implies higher risk for investors and, thus, a higher issue spread <i>ceteris paribus</i> . Results in the literature validate this expectation; see, for example, Gabbi and Sironi (2005) and Binfarè and Gatti (2014).
<i>Ln_Size</i>	The natural logarithm of the bond's size previously converted into American dollars (exchange rate as of 10 September 2015) in the case of issuances in other currencies. The coefficient is expected to be negative since a larger size should, <i>ceteris paribus</i> , imply greater liquidity in the secondary markets and thus less liquidity risk for investors. Such expectation is validated in the literature; see, for example, Gabbi and Sironi (2005) and Binfarè and Gatti (2014).
<i>G3_CRNCY</i>	Dummy variable that takes value 1 if the bond is denominated in a G3 currency (i.e., US dollar, euro, Japanese yen) and 0 otherwise. There is no clear expectation regarding the coefficient, as, on the one hand, G3-denominated bonds should bear less currency risk (i.e., less volatile exchange rates), but on the other hand, the credit rating profile of issuers is generally very different between G3 currencies and non-G3 currencies.
<i>Rank</i>	Dummy variable that takes value 0 if the bond is senior in the payment rank compared to other outstanding debt and 1 if it is subordinated. Subordinated bonds have a lower expected recovery rate in the case of default than senior bonds; accordingly the expected coefficient sign is positive. This risk however, should be already taken into account by the credit rating, hence its statistical significance might be poor (Gabbi and Sironi, 2005).

Issuer characteristics The following variables take into consideration the nature of the issuer and how this impacts the issue spread:

<i>SSA_Dummy</i>	Dummy variable that takes value 1 if the bond is issued by a sovereign, supranational or agency (SSA) and 0 otherwise. This variable should be able to measure the better creditworthiness of the SSAs, which are, by definition, closely linked to national and international governments (if not being part of them). Accordingly, the sign of the coefficient is expected to be negative. This variable is not common in the literature, rather in
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Gabbi and Sironi (2005) a distinction is made between banks and other issuers: following their approach the next variable is included.

Bank_Dummy

Dummy variable that takes value 1 if the bond is issued by a bank and 0 if it is issued by another type of entity. The inclusion of this variable should control for the presence of implicit government guarantees that are not already incorporated into the issuance credit rating. As highlighted above, the use of this variable follows Gabbi and Sironi (2005) where the resulting coefficient is negative and statistically significant.

Market practices The following two independent variables only apply to the green bonds market and are not found elsewhere in the literature: they represent a core feature of this chapter:

Reviewer_Dummy

Dummy variable that takes value 1 if the bond is accompanied by a second party review and 0 otherwise. The expected sign of this coefficient is negative: investors in green bonds should, *ceteris paribus*, reward the assurance provided by the issuer with a lower issue spread.

GBPs_Dummy

Dummy variable that equals 1 if the bond complies with the Green Bond Principles introduced in January 2014, 0 otherwise. There is no clear expectation regarding the sign of the coefficient: issue spread should be lower on average as investors gain increasing comfort with the green bond frameworks but, on the other hand, such increased comfort will favour the issuance of instruments with worse credit standing, thus pushing the coefficient in the positive territory.

Primary market efficiency The following variables are used as proxies for primary market conditions at the moment of issuance:

Bookrunners

This variable represents the number of bookrunners⁷ participating in the bond issuance process. The use of this variable is common in the literature, for example, Gatti et al. (2013). There is no clear expectation on the sign of the coefficient: on the one hand a larger number of bookrunners should be able to collect more interest from investors and thus attain a lower issue spread; on the other hand, the number of bookrunners in the bond issuance management group largely depends on the complexity of the transaction and on market conditions. In fact, as evidenced in Gabbi and Sironi

Syndicate_qual

(2005), this coefficient may change sign according to the time-span under consideration.

This variable measures the prestige of the bookrunners managing the bond issuance and it is constructed by computing the average market share that the bookrunners belonging to the syndicate had in the quarter before issuance. Indeed, in the literature there are some interesting examples showing how bookrunners' reputation may affect equity and debt pricing. For instance, Lin, Pukthuanthong and Walker (2013) argue that the most prestigious investment banks may tend to under-price initial public offerings (IPOs) in order to avoid lawsuits; Gatti et al. (2013) find that issue spread for project finance bonds tend to be lower when prestigious investment banks manage the transaction. Accordingly, there is no expectation regarding the sign of this coefficient as it will largely depend on which of the two effects prevails.

4.2 Sample Statistics and Representativeness

In Table 11.2 key statistics of the sample are presented and compared with the population to understand whether the characteristics of the sample reflect those of the population. Noticeably, the regression sample represents 59 per cent of the population (i.e., 246 out of 417 green bonds issued up to 10 September 2015) with yearly distribution of issuance closely mirrored. Sample representativeness regarding credit rating is an essential feature for generalizing the results derived from our regressions to the entire population given the fundamental role played by this variable in determining the issue spread (Gabbi and Sironi, 2005): the only rating category that is significantly underrepresented in the sample is 'Not Rated/Not Available' but the result is a natural consequence given that the bonds that do not have a rating are more likely to have less disclosure obligations and, hence, their issue spread is less likely to be available from financial data providers such as Bloomberg. Similar evidence comes from considering the size of issuance: the regression sample closely mirrors the population with an exception made for green bonds smaller than USD50 million, as those very small bonds may have less disclosure obligations. The proportions of green bonds issued in G3 currencies and being subordinated in the payment rank are fairly similar in the regression sample and in the population. Furthermore, the representativeness of the sample is very high for the characteristics belonging to the issuer characteristics, market practices and primary market efficiency factors: this is of particular relevance given that in order to answer the two research questions, inference on the population will be conducted based on the coefficients estimated for *Reviewer_Dummy* and *GBPs_Dummy*.

Table 11.2 Selected sample statistics

Factor	Regression Sample			Population	
	Characteristic	% within the sample		% within the population	
	Issued in 2007	0	0	1	0
	Issued in 2008	1	0	1	0
	Issued in 2009	4	2	4	1
	Issued in 2010	30	12	55	13
	Issued in 2011	17	7	31	7
	Issued in 2012	15	6	20	5
	Issued in 2013	31	13	39	9
	Issued in 2014	87	35	129	32
	Issued in 2015	61	25	137	33
Credit rating	<i>Rating_</i> <i>HIGHGRADE</i>	78	32	103	25
	<i>Rating_</i> <i>MEDIUM</i> <i>GRADE</i>	23	9	23	6
	<i>Rating_</i> <i>SUBIG</i>	17	7	17	4
	Not rated	128	52	274	65
	Bond characteristics	Size: < 50 USD Mil.	90	37	233
Size: 50 to 250 USD Mil.		66	27	91	22
Size: 250 to 500 USD Mil.		32	13	34	8
Size: 500 to 1000 USD Mil.		45	18	46	11
Size: > 1000 USD Mil.		13	5	13	3
Currency: G3		127	52	195	47
Rank:		2	1	2	0
Subordinated bonds					
Issuer characteristics		Issuer: SSA	143	58	240
	Issuer: Bank	23	9	51	12
Market practices	Issued with a second party review	165	67	271	65
	Issued after GBPs' introduction	148	60	266	64

Table 11.2 (continued)

Factor	Regression Sample		Population
	Characteristic	% within the sample	% within the population
Primary market efficiency	Bookrunners (average per issuance)	2.43	1.88
	Syndicate market share (average per issuance)	2.56%	2.76%

Note: This table shows selected characteristics of the sample and of the population. Data is sorted according to the factor it belongs to in the regression analysis (left-most column).

4.3 Results

As presented in Table 11.3, there are three estimated OLS regression models: a full model – that is, containing all the candidate variables; a model where the rating dummies are not included; and a model where independent variables are chosen to minimize the Akaike information criterion⁸ (AIC).

Overall, model 1 (the full model) and model 3 (the AIC model) perform fairly similar in terms of adjusted R^2 and indeed use the same independent variables except for *Tenor*, which is not included in model 3. Model 2 (the one with no rating dummies) does not come close to the other two models.

More specifically, the three rating dummies (*Rating_HIGHGRADE*, *Rating_MEDIUMGRADE* and *Rating_SUBIG*) are all significant at a 95 per cent confidence level in model 1 and model 3 and their coefficients' signs behave as expected: compared to the green bonds that do not have a Bloomberg Composite Rating, having a 'high-grade' rating, that is, within the [AAA, AA–] range, implies an issue spread that is on average 75.21 (71.39) basis points lower *ceteris paribus*. Similarly, having a 'medium-grade' rating, that is, within the [A+, BBB–] range, implies an issue spread that is on average 85.42 (84.69 in model 3) basis points lower *ceteris paribus*. However, having a 'sub-investment grade' rating, that is, BB+ or below, entails an issue spread that is on average 213.41 (212.65 in model 3) basis points higher *ceteris paribus*. These results are in line with the relevant literature, in particular Gabbi and Sironi (2005) and Buscaino et al. (2012).

The OLS coefficient for the independent variable *Tenor* is not statistically significant in model 1 or in model 2; accordingly, statistical inference on the population is not conducted.

$\ln_size$'s coefficients are negative in all three models as expected and statistically different from zero, with a p -value lower than 10 per cent in model 1, lower than 1 per cent in model 2 and lower than 5 per cent in model 3. This being a logarithmic transformation, the coefficient is interpreted as follows: on average, as the size of the green bond changes by p per cent, the issue spread changes by $\beta * \ln \frac{100+p}{100}$ *ceteris paribus*. Accordingly, as the size

Table 11.3 Regression analysis output

Independent Variables	Model 1 (Full Model)			Model 2 (No Rating Dummies)			Model 3 (AIC Method)		
	Coeff.	p-value	S.p. R^2	Coeff.	p-value	S.p. R^2	Coeff.	p-value	S.p. R^2
Intercept	300.10	***		510.73	***		321.56	***	
Rating_HIGHGRADE	-75.21	***	2.68%				-71.39	***	2.58%
Rating_MEDIUMGRADE	-85.42	**	1.27%				-84.69	**	1.25%
Rating_SUBIG	213.41	***	5.59%				212.65	***	5.55%
Tenor	1.57		0.11%	-1.21		0.07%			
Ln_size	-11.33	*	0.71%	-20.12	***	2.39%	-12.04	**	0.83%
G3_CRNCY	46.59	**	1.33%	54.63	***	1.87%	48.24	***	1.44%
Rank	214.90	**	1.16%	393.54	***	4.28%	215.57	**	1.17%
SSA_Dummy	-63.49	***	1.35%	-125.66	***	7.23%	-65.30	***	1.44%
Bank_Dummy	-56.25	*	0.65%	-113.66	***	2.86%	-59.03	*	0.73%
Reviewer_Dummy	-51.92	***	1.81%	-75.25	***	3.9%	-54.17	***	2.03%
GBPs_Dummy	35.38	*	0.7%	23.37		0.31%	36.67	*	0.75%
Bookrunners	7.74	*	0.59%	11.20	**	1.46%	7.91	*	0.62%
Syndicate_qual	5.12		0.36%	5.65		0.46%	5.49		0.42%
R^2	54.19%			40.57%			54.08%		
Adjusted- R^2	51.62%			38.04%			51.72%		

Note: This table contains the estimated OLS coefficients, p -values and semi-partial R^2 for the three specifications of the regression model used in the regression analysis.

of the green bond increases by 1 per cent, the issue spread decreases on average by 0.1133 basis points *ceteris paribus* (0.2012 and 0.1204 according to model 2 and 3 estimates).

G3_CRNCY has positive and statistically significant coefficients in all of the three models (46.59, 54.63, 48.24), which means that, on average, green bonds issued in USD, EUR and JPY have issue spreads higher than those issued in other currencies *ceteris paribus*. The result may seem counterintuitive as the G3 currencies bonds are expected to bear virtually no currency risk compared to emerging market currencies and benefit from greater liquidity in the secondary markets. However, a possible explanation is that non-G3 currencies markets are open only to high-grade rated issuers that are able to issue at a lower spread compared to lower-rated issuers.

Rank's coefficient is positive as expected and statistically significant in all three models, meaning that, on average, subordinated bonds have an issue spread that is 214.90 (215.57 in model 3) basis points higher than unsubordinated bonds *ceteris paribus*. Notably, in model 2, the coefficient has greater magnitude (393.54) given the correlation that exists between payment rank and credit rating of a bond. This result is in line with previous authors, especially Gabbi and Sironi (2005).

SSA_Dummy shows that, on average, SSA green bonds' issue spreads have been 63.49 basis points lower *ceteris paribus* (65.30 in model 3); similar to that noticed for *Rank*, omitting the credit rating dummies in model 2 leads to an almost double coefficient (125.66). The sign of the *SSA_Dummy* variable is understandable and relates to the fact that SSAs are closely linked to national and international governments (if not being part of them): this fact benefits their credit standing and thus lowers their issue spreads.

Following the same reasoning as Gabbi and Sironi (2005) the regression model also includes the variable *Bank_Dummy* to control for the presence of implicit government guarantees, such as 'too-big-to-fail', which may not have already been incorporated in the issue rating. In line with the previous literature, the coefficient is negative in all of the three models but with a *p*-value lower than 10 per cent in model 1 and model 3 and lower than 1 per cent in model 2, pointing to an issue spread that is, on average, 56.25 (113.66 in model 2, 59.03 in model 3) basis points lower for green bonds issued by banks with respect to non-bank issuers *ceteris paribus*.

The *Reviewer_Dummy* coefficient highlights a very important result for the purpose of this chapter: on average, *ceteris paribus*, the issue spreads of the green bonds that have a second party review are 51.92 (75.25 in model 2, 54.17 in model 3) basis points lower than those that do not have one, with a *p*-value always lower than 1 per cent. Accordingly, on the basis of the OLS regression analysis, the research question 'Are the issue spreads of green bonds with a second party review lower than those without one?' is answered affirmatively.

Just as important for this chapter is the coefficient for *GBPs_Dummy*, which serves the purpose of answering the research question 'Are green bonds' issue spreads decreasing in response to the introduction of the Green Bond Principles?' Indeed, even if the coefficient indicates that issue spreads are on average 35.38 (23.37 in model 2, 36.67 in model 3) basis points higher, *ceteris paribus*, since the introduction of the Green Bond Principles, the research question will not be promptly answered negatively for at least two reasons. First, the coefficient is not statistically different from zero in model 2 (nor is it at the 95 per cent confidence level in models 1 and 3); second, the positive sign of the coefficient may be indicating something very interesting: the Green Bond Principles are working as a catalyst

for green bond market development in such a way that ‘riskier’ bonds are now able to enter the market even if they have to pay a certain cost in terms of issue spread. This intuition is tested using a *t*-test for the mean difference in two samples assuming unequal variances.⁹ Table 11.4 shows the results of the test for the regression sample (left-hand side) and the population (right-hand side): the null hypothesis that the mean credit rating is the same before and after the introduction of the Green Bond Principles is rejected (*p*-value = 0.0000), and in particular the average credit rating is higher (i.e., better) before the introduction of the Green Bond Principles. This empirical result is even more evident when the rating distributions before and after the introduction of the Green Bond Principles are considered for both the regression sample and the population, as shown in the bottom part of Table 11.4: sub-investment-grade green bonds and almost all of the medium-grade green bonds were issued after the introduction of the Green Bond Principles.

The *Bookrunners* variable has a positive coefficient (7.74 in model 1, 11.2 in model 2 and 7.91 in model 3) but displays a *p*-value lower than 5 per cent only in model 2. The result is somewhat counterintuitive as a larger syndicate is expected to reach a larger investor base and, thus, attain a lower issue spread; however, as highlighted already in Gabbi and Sironi (2005) it may occur that the coefficient turns positive as higher uncertainty in the market requires issuers to hire a larger syndicate to launch their bonds.

Similarly, *Syndicate_qual* is not statistically different from zero in all of the three models, which explains how the positive coefficient of 5.12 (5.65 in model 2 and 5.49 in model 3) is not in line with the results of previous studies – for example, Gatti et al. (2013). Another possible explanation is how the *Syndicate_qual* variable was constructed: the market shares retrieved from Bloomberg refer to the ‘international bonds’ category. There are some very active investment banks that have very low market shares in the international bonds league table but are market leaders in the green bonds space (e.g., SEB, Incapital).

5 CONCLUSIONS

The green bond market is acknowledged as one of the fastest-growing areas in the financial markets: a USD72 billion market that experienced a 165 per cent annualized growth rate during the past eight years. Such exceptional development would not have been possible without the interest shown by both investors and issuers, whose push for increasing standardization and continuous innovation has led to several types of green bonds entering the market and even more opportunities to invest in and finance green projects. Market participants, in particular consultancy firms and financial advisors, welcomed these recent developments as fundamental steps in order to strengthen investors’ confidence as well as to facilitate future green bonds issuances. Unfortunately, though, no formal study assessing the benefits of these advancements has been conducted to date. Building on this acknowledgement, the main objective of this chapter was to quantitatively measure the benefits to issuers originating from the increasing standardization occurring in the market (i.e., the Green Bond Principles) and from a specific innovation (i.e., the second party review).

In particular, this chapter posed two research questions: (1) Are green bonds’ issue spreads decreasing in response to the introduction of the Green Bond Principles? (2) Are the issue spreads of green bonds with a second party review lower than those without

Table 11.4 T-test for the mean difference in credit rating, pre- and post-GBPs

Regression Sample		Population			
T-test: Two-sample assuming unequal variances		T-test: Two-sample assuming unequal variances			
	Pre-GBPs	Post-GBPs			
Mean	19.51	15.43	Mean	19.61	16.09
Variance	18.67	27.47	Variance	15.02	25.64
Observations	37	81	Observations	47	96
Hypothesized mean difference	0		Hypothesized mean difference	0	
df	101		df	116	
t-stat.	65.38		t-stat.	64.42	
p-value one-tail	0.0000		p-value one-tail	0.0000	
p-value two-tails	0.0000		p-value two-tails	0.0000	
Rating distribution		Rating distribution			
Rating_HIGHGRADE	35%	30%	Rating_HIGHGRADE	29%	22%
Rating_MEDIUMGRADE	3%	14%	Rating_MEDIUMGRADE	2%	8%
Rating_SUBIG	0%	11%	Rating_SUBIG	0%	6%
Rating_NR/NA	62%	45%	Rating_NR/NA	69%	64%

Note: This table shows how the rating distribution is influenced by the introduction of GBPs; this evidence is statistically proven with a t-test for the mean difference in credit rating (top part of the table) and qualitatively gauged by looking at the rating distributions (bottom part of the table).

one? These questions were addressed by means of a regression analysis: the issue spread over mid-swap rate, the dependent variable, was modelled through several independent variables that were methodically chosen, also taking into account previous studies that enquired into the issue spread determinants for corporate, financial institutions and project finance bonds.

The research question ‘Are green bonds’ issue spreads decreasing in response to the introduction of the Green Bond Principles?’ is answered negatively. In fact, according to a first interpretation of the regression analysis, issue spreads are on average, *ceteris paribus*, higher after the introduction of the Green Bond Principles. However, this result needs a closer look: in response to the introduction of the principles, many bonds with a worse credit rating have been able to enter the market and, thus, have influenced the coefficient in the regression. This interesting conclusion was empirically tested with a *t*-test for the mean difference in credit rating and will open to the possibility of further studies, especially as the number of observations that can be included in the analysis is bound to increase as new green bonds are issued in the market.

The research question ‘Are the issue spreads of green bonds with a second party review lower than those without one?’ is answered affirmatively. According to this chapter a green bond accompanied by a second party review has on average, *ceteris paribus*, a lower issue spread. This conclusion highlights a very important element of the green bond marketplace: investors appreciate those issuers that are committed to green financing and do factor such commitment into their investment evaluation process.

NOTES

1. The market prices newly issued bonds with an ‘issue spread’, which adds to a benchmark rate: this means that a higher issue spread over the benchmark rate leads to a higher cost of issuing the bond for the issuer (either in the form of higher coupon rate or greater discount to par).
2. Leadership in Energy and Environmental Design (LEED) is a set of rating systems developed by the US Green Building Council regarding the design, construction, operation, and maintenance of green buildings, homes and neighbourhoods.
3. Energy Star is an international standard for energy-efficient consumer products, originating in the United States in 1992 and currently adopted in Australia, Canada, Japan, New Zealand, Taiwan and in the European Union.
4. Building Research Establishment Environmental Assessment Method (BREEAM) is the world’s longest established method of assessing, rating and certifying the sustainability of buildings, first introduced in 1990.
5. The G3 currencies are American dollar, euro, Japanese yen.
6. Second party reviewers are entities with various backgrounds, mostly ESG (environmental, social and governance) research houses and scientific standards groups. As of today, the most active second party reviewers are: Center for International Climate and Environmental Research (CICERO), Vigeo Eiris Rating, DNV-GL, Oekom, Sustainalytics, SUST4IN and KPMG.
7. ‘Bookrunners’ are the investment banks in the bond issuance management group that ‘run the books’ – that is, record and manage the orders by investors and allocate the bonds at the moment of issuance (Rosenbaum and Pearl, 2013).
8. $AIC = 2k - 2\ln(L)$, where k is the number of estimated parameters in the model and L is the maximum value of the likelihood function in the model. AIC rewards goodness of fit (as assessed by the likelihood function), but it also includes a penalty that is an increasing function of the number of estimated parameters. Given a set of candidate models for the data, the preferred model is the one with the minimum AIC value.
9. In order to complete the test, the Bloomberg Composite credit rating for each bond was converted into a number, AAA = 20, AA+ = 19, AA = 18, AA- = 17, A+ = 16, A = 15, A- = 14, BBB+ = 13, BBB = 12, BBB- = 11, BB+ = 10, BB = 9, BB- = 8, B+ = 7, B = 6, B- = 5, CCC+ = 4, CCC = 3, CCC- = 2, C = 1, D = 0; Not Rated/Not Available were not considered.

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